

Multi-objective Optimization of Power Supply and Demand Balance Considering the Impact of Coal Prices on Coal-Fired Power Generation Willingness

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Abstract—Under the background of the construction of new power systems, traditional researches mostly consider the impact of source and load uncertainties such as wind, solar, water, and load on operation risks, while ignoring the impact of coal prices. This work proposes a multi-objective optimization method for power supply and demand balance that considers the impact of coal prices on coal-fired power generation willingness. Firstly, indicators such as the cost factor and efficiency factor of coal-fired power generation are defined. And based on this, a method for constructing the willingness curve of coal-fired power generation is proposed. Furthermore, by combining the construction method of source and load scenario set considering the spatiotemporal correlation, the quantitative analysis of power supply risk is achieved. Then, a multi-objective optimization method for power supply and demand balance is proposed, considering expected daily operating costs, inter-region transmission electricity, and electricity shortage, as well as constraints such as coal-fired power generation willingness and equipment operation. Finally, the method is applied to the test system to analyze the impact of different coal prices on system operational economy, self-balancing degree, and operational risk.

Keywords—Willingness for coal-fired power generation, power system, supply and demand balance, multi-objective optimization, electricity shortage

I. INTRODUCTION

Under the "dual carbon" goal, the construction of new power systems has become a key means to solve environmental pollution and resource shortages in the energy sector [1]. To cope with the significant uncertainty brought by the high proportion of renewable energy access, how to analyze and

consider the risk transmission of primary energy such as coal, wind, solar, and water has become the key to optimizing the supply-demand balance of the new power system [2].

At present, most primary energy risk analyses focus on uncertain renewable energy sources such as wind, solar, and water. Reference [3] considers modeling the degree of load shedding of wind turbines (WT) on uncertain factors such as wind speed and output. Reference [4] proposes an improved interval optimization method to simulate the uncertainty of inflow runoff and photovoltaic (PV) power output in cascade hydropower stations. Reference [5] considers the regional, temporal, temporal autocorrelation, and cross-correlation of source and load in each area, and establishes a set of uncertain operating scenarios such as wind, illuminance, and load.

In terms of optimizing the electricity balance between supply and demand, Reference [6] proposes a generation transmission energy storage collaborative optimization model that considers the seasonal imbalance risk caused by the long-term uncertainty of renewable energy generation in the power system. Reference [7] proposes a power system planning model based on long-term distributed robust optimization, which minimizes the total investment and operating costs, as well as the expected penalty costs for load shedding and wind power curtailment. Reference [8] proposes a multi-type flexible resource coordination method for power system expansion planning using technologies such as thermal power flexible transformation, energy storage system, demand side response, and centralized solar power plants.

Most of the traditional research mentioned above focuses on the risks brought by uncertain sources such as wind, illuminance, water, and load, and regards coal-fired power units (CP) as

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deterministic power sources that meet operational constraints. However, in September 2021, under extreme coal prices, coal-fired power companies suffered cash flow losses, and the "planned electricity market coal" contradiction erupted. In some areas, coal-fired power companies had tight coal inventories and decreased willingness to generate electricity, resulting in a large shortage of electricity supply [10]. Therefore, the indirect impact of coal prices on the risk of electricity supply cannot be ignored. In addition, traditional research on optimizing power supply and demand balance has not comprehensively considered multiple indicators such as system operation economy, self-balancing degree, and risk. Regarding the above issues, the work is as follows:

- A method for constructing the willingness curve of CP is proposed, which can quantitatively analyze the indirect impact of coal prices on the willingness of CP to generate electricity.
- A multi-objective optimization method for power supply and demand balance considering the impact of coal prices on coal-fired power generation willingness is proposed. This method simultaneously considers the uncertainty of source load and the impact of coal prices on the risk of power supply guarantee, while also consider multiple indicators such as expected daily operating costs, interval transmission capacity, and power shortage.

II. ANALYSIS OF RISK FACTORS IN POWER SUPPLY

A. Analysis of willingness for CP generation

This section mainly aims to quantitatively analyze the impact of coal prices on the willingness to generate CP. Intention is essentially an upper limit, and the willingness for CP generation can be reflected in the optimization of power supply and demand balance through the upper limit of power generation. The CP generation willingness curve is defined as the mapping relationship between the cost factor and the willingness factor of power generation. Among them, the generation cost factor of CP λ_b is defined as the ratio of unit generation cost c_{cc} to unit grid price c_{cb} , reflecting the benefits of coal power generation. The power generation willingness factor λ_w is defined as the ratio of monthly power generation to installed capacity in high coal price scenarios, reflecting the willingness to generate electricity from CP:

$$\begin{cases} \lambda_b = c_{cc} / c_{cb}, & c_{cc} = c_c / p_c \\ \lambda_w = P_g / S_c \end{cases} \quad (1)$$

where, c_c represents the unit coal price. p_c is the unit of coal that can generate electricity.

Fig. 1 shows a schematic diagram of the willingness curve for CP generation. In the scenario of low coal prices, the output of CP is mainly affected by factors such as load, and the upper limit is mainly constrained by equipment operation. When the coal price reaches a certain threshold, the revenue from CP generation decreases, and CP output is not only constrained by equipment operation, but also by the willingness to generate

electricity. The Spearman coefficient can be used to analyze the correlation between the historical data of CP generation cost factors and generation willingness factors, and then select high coal price historical scenarios to determine the threshold for high coal price scenarios.

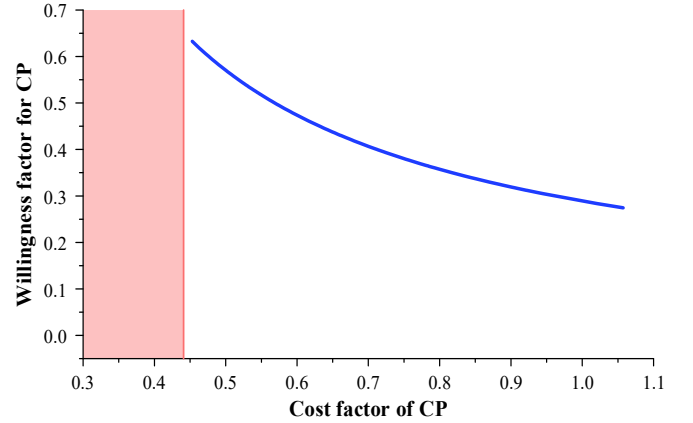


Fig. 1. Schematic diagram of CP generation willingness curve

B. Scene set construction method for WT, PV, and load power

This work adopts a method of constructing WT, PV, and load power scene sets that considers the spatiotemporal correlation of source and load. The features considered include temporal, regional, temporal autocorrelation, and cross-correlation between sources and loads in different regions [5]. The specific process is: 1) Preprocess historical source load data day by day and hour by hour. 2) Calculate the temporal probability distribution parameters of various source and load in each area based on maximum likelihood estimation method. 3) Based on Spearman correlation coefficient method, analyze the autocorrelation of historical data of various sources and loads in each area and the cross-correlation between sources and loads in each area. 4) Construct source and load scenes with target autocorrelation for each region based on Latin hypercube sampling-Cholesky decomposition method. 5) Based on particle swarm optimization algorithm, select a set of source and load scenes with target cross-correlation.

III. OPTIMIZATION METHOD FOR POWER SUPPLY AND DEMAND BALANCE CONSIDERING THE IMPACT OF CP ON COAL-FIRED POWER GENERATION WILLINGNESS

A. Optimization objectives

The economic efficiency of power system operation is a major focus of optimization. Minimizing power shortages as much as possible is also a core element in optimizing the balance between power supply and demand. In this regard, this work selects the expected daily power generation cost and the expected daily power shortage as the objectives. In addition, to ensure the self-balance of each area as much as possible, this work also considers the expected daily interval transmission power as an optimization variable. The specific expression of the optimization objective function is:

$$\begin{cases} \min f_1 = \sum_{d=1}^{sc} \sum_{t=1}^{24} (c_{cc} P_{CP}^{d,t} + c_{ch} P_{HT}^{d,t} + c_{cw} P_{WT}^{d,t} + c_{cp} P_{PV}^{d,t}) / sc \\ \min f_2 = \sum_{d=1}^{sc} \sum_{t=1}^{24} (P_L^{d,t} - P_{CP}^{d,t} - P_{HT}^{d,t} - P_{WT}^{d,t} - P_{PV}^{d,t}) / sc \\ \min f_3 = \sum_{d=1}^{sc} \sum_{t=1}^{24} \sum_{i=1}^{area} \sum_{j=1}^{area} (P_{i,j}^{d,t}) / sc, i \neq j \end{cases} \quad (2)$$

where, c_{cc} , c_{ch} , c_{cw} , and c_{cp} are the unit power generation costs of CT, hydraulic turbines (HT), WT, and PV, respectively. sc and $area$ represents the number of scenes and the number of regions, respectively. $P_{CP}^{d,t}$, $P_{HT}^{d,t}$, $P_{WT}^{d,t}$, and $P_{PV}^{d,t}$ represent the power generation of CP, HT, WT, and PV at time t in scenario d . $P_L^{d,t}$ is the electrical load at time t in scenario d , and $P_{i,j}^{d,t}$ is the transmission power between region i and j at time t in scenario d .

B. Constraint condition

The optimization of power supply and demand balance mainly involves operational constraints on power generation equipment such as CP, non-adjustable hydraulic turbines (AHT), non-adjustable hydraulic turbines (NHT), WT, PV, energy storage (ES), pumped storage (PS), etc. Among them, NHT, WT, and PV are uncertain power sources, and their output can be determined by the scenario set constructed in Section II. In addition to the upper and lower limits of equipment and climbing constraints, CP operation constraints also need to consider the impact of coal price on the willingness of coal power generation in Section II, specifically expressed as:

$$\begin{cases} K_{CP}^{\min} S_{CP}^a \leq P_{CP}^{d,t,a} \leq K_{CP}^{\max} S_{CP}^a \\ |P_{CP}^{d,t+1,a} - P_{CP}^{d,t,a}| \leq K_{pp} S_{CP}^a \\ P_{CP}^{d,t,a} \leq \lambda_b S_{CP}^a, \text{ if } \lambda_b > \lambda_b^T \end{cases} \quad (3)$$

where, $P_{CP}^{d,t,a}$ represents the CP output of the a -th region at the t -th moment of the d -th scene. S_{CP}^a is the installed capacity of CP in region a . $|\cdot|$ represents the absolute value of an element.

$K_{CT}^{\max}$ and $K_{CT}^{\min}$ are the maximum and minimum output coefficients of CP, respectively. K_{pp} is the climbing rate constraint coefficient. λ_b^T is the power generation cost factor corresponding to the threshold of high coal price scenarios. The last equation reflects the constraint of coal prices on CP power output, which is satisfied in high coal price scenarios.

AHT needs to meet the upper and lower limits of operation constraints, which are specifically expressed as:

$$K_{AHT}^{\min} S_{AHT}^a \leq P_{AHT}^{d,t,a} \leq K_{AHT}^{\max} S_{AHT}^a \quad (4)$$

where, $P_{AHT}^{d,t,a}$ represents the AHT output of the a -th region at the t -th moment of the d -th scene. S_{AHT}^a is the installed capacity of CP in region a . $K_{AHT}^{\max}$ and $K_{AHT}^{\min}$ are the maximum and minimum output coefficients of AHT, respectively.

The constraints on PS operation can be expressed as:

$$\begin{cases} \sum_{t=1}^{24} P_{d,t,a}^{PS,in} \eta_{in}^{PS} = \sum_{t=1}^{24} P_{d,t,a}^{PS,out} / \eta_{out}^{PS} \\ 0 \leq P_{d,t,a}^{PS,in} \leq x_{d,t,a}^{PS,in} P_{in,max}^{PS} \\ 0 \leq P_{d,t,a}^{PS,out} \leq x_{d,t,a}^{PS,out} P_{out,max}^{PS} \\ x_{d,t,a}^{PS,in} + x_{d,t,a}^{PS,out} \leq 1 \end{cases} \quad (5)$$

where, $P_{d,t,a}^{PS,in}$ and $P_{d,t,a}^{PS,out}$ represent the charging and discharging power of the PS in the a -th region at the t -th moment of the d -th scene, respectively. η_{in}^{PS} and η_{out}^{PS} represents the charging and discharging efficiency of PS. $P_{in,max}^{PS}$ and $P_{out,max}^{PS}$ represent the maximum charging and discharging power of PS. $x_{d,t,a}^{PS,in}$ and $x_{d,t,a}^{PS,out}$ represent the 0-1 variables that characterize the charging and discharging status of the PS in the a -th region at the t -th moment of the d -th scene.

The constraints on ES operation can be expressed as:

$$\begin{cases} R_{d,t+1,a}^{ES} = (1 - \eta_l^{ES}) R_{d,t,a}^{ES} + P_{d,t,a}^{ES,in} \eta_{in}^{ES} - P_{d,t,a}^{ES,out} / \eta_{out}^{ES} \\ 0 \leq P_{d,t,a}^{ES,in} \leq x_{d,t,a}^{ES,in} P_{in,max}^{ES} \\ 0 \leq P_{d,t,a}^{ES,out} \leq x_{d,t,a}^{ES,out} P_{out,max}^{ES} \\ x_{d,t,a}^{ES,in} + x_{d,t,a}^{ES,out} \leq 1 \\ R_{a,min}^{ES} \leq R_{d,t,a}^{ES} \leq R_{a,max}^{ES} \\ K_{ES} R_a^{ES,max} = R_{d,s,a}^{ES} = R_{d,e,a}^{ES} \end{cases} \quad (6)$$

where, η_{in}^{ES} , η_{out}^{ES} , and η_l^{ES} represent the charging efficiency, discharging efficiency, and loss coefficient of ES, respectively. $P_{d,t,a}^{ES,in}$, $P_{d,t,a}^{ES,out}$ and $R_{d,t,a}^{ES}$ respectively represent the charging power, discharging power, and storage capacity of the ES in the region a at time t in the d -th scenario. $P_{in,max}^{ES}$ and $P_{out,max}^{ES}$ represents the maximum charging and discharging power of ES. $x_{d,t,a}^{ES,in}$ and $x_{d,t,a}^{ES,out}$ are the 0-1 variables that represent the charging and discharging status of ES in the region a at time t in scene d . $R_{a,max}^{ES}$ and $R_{a,min}^{ES}$ represent the maximum and minimum capacity of the ES in region a . K_{ES} is the initial output coefficient of ES. $R_{d,s,a}^{ES}$ and $R_{d,e,a}^{ES}$ represent the initial and final ES of the d -th scene in the a -th area.

Power transmission between regions must meet the upper limit constraint of the interconnection line:

$$0 \leq P_{i,j}^{d,t} \leq P_{i,j}^{\max}, i \neq j \quad (7)$$

where, $P_{i,j}^{\max}$ represents the maximum allowable transmission power between regions i and j .

C. Solution method

In response to the multi-objective optimization model of power supply and demand balance considering the impact of coal prices on CP generation willingness, a fuzzy membership function is used to transform the three multi-objective objectives

into a single objective, and commercial solvers such as CPLEX and GUROBI are used to solve the single objective optimization problem.

IV. CASE STUDY

A. Analysis of the Willingness Curve for CP Generation

This work applies the optimization method of power supply and demand balance considering the impact of coal prices on CP generation willingness to H province in China. Historical data such as coal prices, monthly power generation, electricity prices, and installed capacity from 13 CP plants in the province are selected for analysis. The willingness factor based on monthly power generation and installed capacity is calculated. The cost factor by comparing the unit power generation cost with the grid electricity price is calculated. According to the Spearman coefficient analysis, there is a strong negative correlation between the willingness factor and cost factor historical data from September to December 2021, which is consistent with the actual situation of high coal prices and decreased willingness of CP plants to generate electricity at that time. Historical data from 10 CP plants with Spearman coefficients below -0.8 are selected to construct a CP generation willingness curve. Selecting polynomials, exponents, logarithms, power functions, and S-shaped functions to fit the CP generation intention curve, the error indicators of each fitting curve are shown in **Table 1**. The RMSE under the power function is the lowest [11].

TABLE I. ERROR UNDER DIFFERENT FITTING FUNCTIONS

Function	Polynomial	Exponential	Logarithm	Power	S
RMSE	0.0914	0.0909	0.0908	0.0905	0.0918

The power function is selected to construct the intention curve for CP generation, as shown in **Fig. 2**. As the cost factor of CP generation gradually increases, the willingness factor to generate electricity gradually decreases. According to the correlation analysis based on Spearman correlation coefficient, the cost factor threshold corresponding to the high coal price

scenario is 0.61, and the coal price is about 1300 yuan/ton at this time. The historical data from January to August 2021 basically did not reach the threshold, while the values from September to December were basically above the threshold. In the high coal price scenario, the on-grid electricity price of CP units is basically reported at full capacity to seek maximum benefits, that is, a 20% increase on the benchmark electricity price. At this time, the cost factor of power generation mainly depends on the unit coal price.

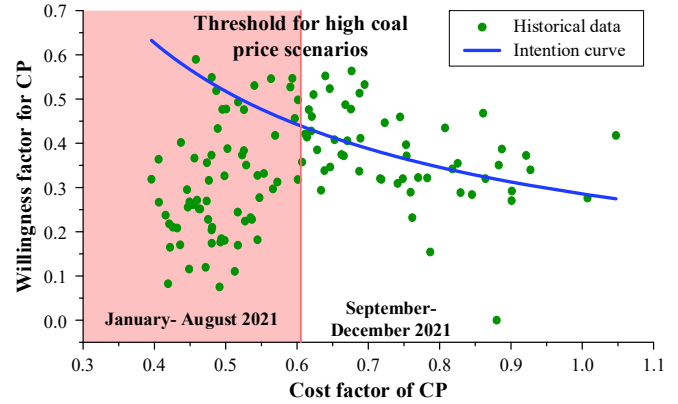


Fig. 2. Willingness curve for CP generation in the case study

B. Optimization analysis of power supply and demand balance considering the impact of coal prices on CP generation willingness

The method in this work is applied to H province to analyze the power shortage situation under different coal prices. **Table 2** shows three optimization objectives: expected daily power generation cost, expected daily power shortage, and expected daily interval transmission power under coal prices ranging from 1200 to 1900 yuan/ton. The changing trends of the three objectives are shown in **Fig. 3**.

TABLE II. OPTIMIZATION RESULTS OF POWER SUPPLY AND DEMAND BALANCE UNDER DIFFERENT COAL PRICE SCENARIOS

Coal price	1200	1300	1400	1500	1600	1700	1800	1900
Electricity shortage (10^4 kWh)	0	414.1	974.1	1370.6	1447.6	1819.8	2239.1	2543.5
Interval transmission of electricity (10^4 kWh)	7883.4	11308.4	13561.4	13200.9	11700.4	11857.9	12123.3	12027.4
Operating cost (yuan)	16288.3	16076.2	16055.0	15853.6	15635.8	15485.0	15337.2	15160.2

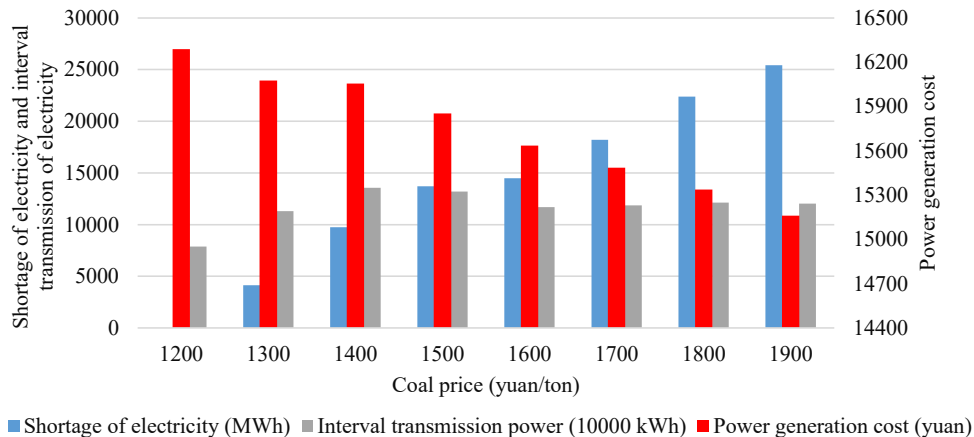


Fig. 3. Trend of optimization objectives under different coal price scenarios

As shown in Fig. 3, with the gradual increase of unit coal price, the expected daily shortage of electricity gradually increases, and the expected daily power generation cost gradually decreases. The coal price of 1200 yuan/ton has not reached the threshold for high coal price scenarios, and there is no expectation of power shortage at this time. The output of CP is relatively large, and the daily power generation cost of the system is high. When the coal price reaches the threshold of high coal price scenario at 1300 yuan/ton, the willingness of CP units to generate electricity decreases, and the system experiences a power shortage. As coal prices continue to increase, the upper limit of CP output decreases, the shortage of electricity gradually

increases, and the cost of power generation also decreases with the decrease in CP output. Before the coal price reaches the threshold, the output of CP units in each zone is relatively high, and the amount of transmitted electricity in each zone is relatively small. In high coal price scenarios, CP units have smaller output. In order to minimize power shortages and shortages, interval resource allocation and complementary coordination are used to ensure the safe and stable operation of the system as much as possible. Therefore, compared to before reaching the threshold, the amount of transmitted electricity in the interval increases after reaching the high coal price scenario threshold.

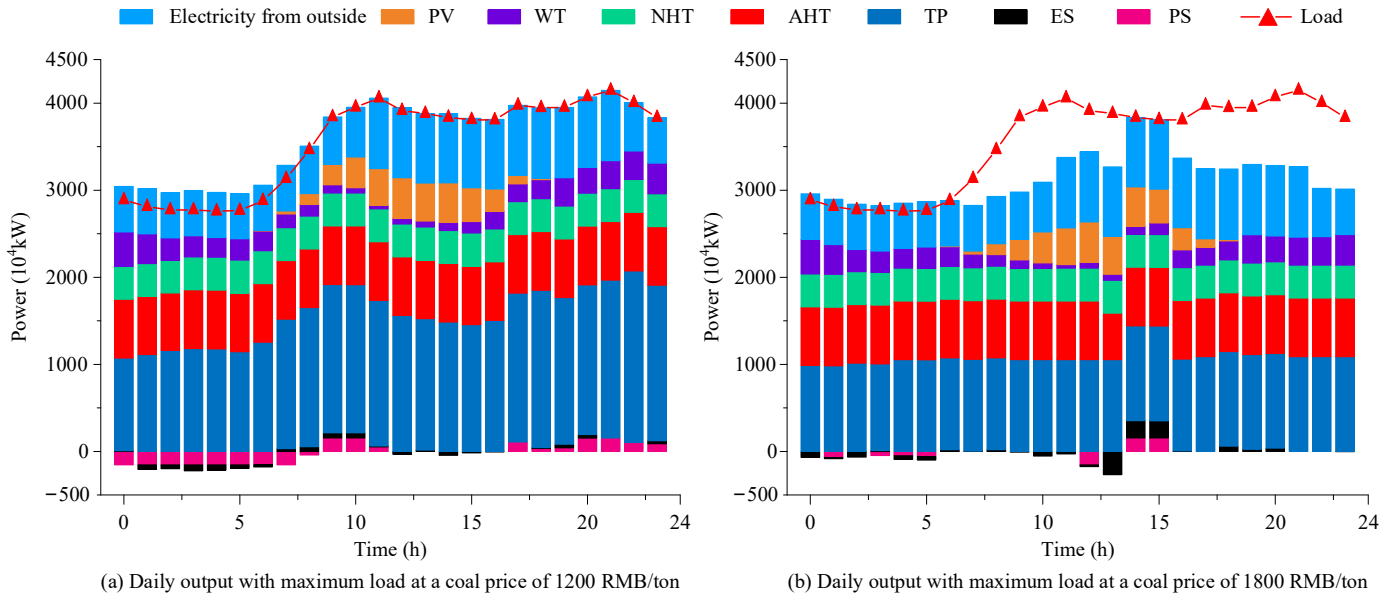


Fig. 4. System operation status during the maximum load day under different coal prices

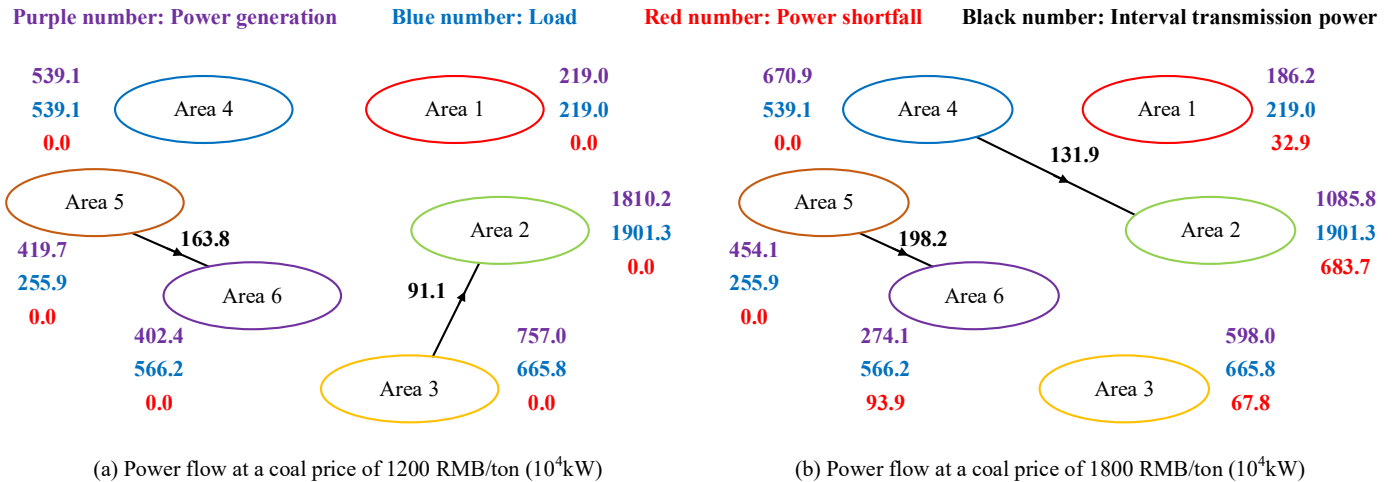


Fig. 5. Power flow distribution at maximum load time under different coal prices

Fig. 4 shows the daily output of each power source in the system under typical scenarios of maximum load when coal prices are 1200 and 1800 yuan/ton. When the coal price is 1200 yuan/ton, the trend of CP output and load changes is similar, with higher output during the noon and evening peak hours and lower output at night. ES and PS charge at night and discharge

at their peak during the noon and evening rush hours. When the coal price reaches 1800 yuan/ton, the willingness for CP generation decreases and the output of CP units decreases. The system experiences a power shortage during most of the day. At night, due to the decrease in power output, the charging capacity of ES and PS decreases, and the peak capacity also decreases.

Taking into account factors such as self-loss of electricity, ES and PS discharge at 14-15 hours, and there is no power shortage at that time. The remaining power of PS is released during the evening rush hour to alleviate the pressure of electricity consumption during the evening rush hour. **Fig. 5** shows the distribution of system power flow at the moment of maximum load under different coal prices. When the coal price is 1200 yuan/ton, there is no electricity shortage in each area, and the two regions have achieved self-balancing. When the coal price reaches 1800 yuan/ton, there will be a power shortage in the four major areas, and Region 2 will experience a significant power shortage due to high load. In addition to meeting self-balancing requirements, Regions 4 and 5 also deliver power to regions 2 and 6 respectively to reduce the power gap in each area.

V. CONCLUSION

Traditional research mostly considers the impact of source and load uncertainty factors on the risk of power supply guarantee, while the impact of coal prices cannot be ignored. This work proposes a multi-objective optimization method for power supply and demand balance that considers the impact of coal prices on CP generation willingness. The conclusions are as follows:

1. When the coal price is below the threshold of high coal price scenarios, only the impact of coal price on system operating costs is considered. When reaching a certain threshold, it is also necessary to consider the impact of coal prices on the willingness to generate coal-fired power. And as coal prices gradually increase, the willingness of CP gradually decreases.
2. This work comprehensively considers the impact of coal prices on the willingness of CP to generate electricity, as well as multiple factors affecting power supply guarantee risks such as source and load uncertainty. It also takes into account multiple indicators such as system operation economy, self-balancing degree, and power shortage risk.
3. When the coal price reaches the threshold for high coal price scenarios, considering its impact on the willingness of CP to generate electricity, there may be a power gap, and the amount of transmitted electricity in the interval will also increase to reduce the power shortage. As coal prices increase, the willingness to generate electricity from coal-fired power decreases, the amount of electricity generated decreases, the operating costs of the system decrease, but the shortage of electricity increases.

Subsequently, research on power system power planning considering the impact of coal prices on CP generation willingness will be conducted. Simultaneously studying the improvement of power gap through flexible regulation resources such as small hydropower, demand side response, and electric vehicles.

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